MULTI-AGENT LLM EVALUATION × INTERPRETABILITY

Same game.
Different words.
Different behavior.

Audited pilots of strategic safety, context, and SAE steering

EXPLORATORY PILOT

Qwen2.5-7B-Instruct F16 - two H100 lanes

The strongest result is a validity warning.

EVIDENCE IN ONE SLIDE

SURFACE

8.4–89.2%

Unsafe-rate span across 18 prompt variants

CONTEXT

+34.0 pp

largest T=0 live effect vs abstract

COMPREHENSION

FAILED

12.5% state update; 17.2% terminal score

ROBUST OUTPUT DIFFERENCES $\neq$ VERIFIED UTILITY OPTIMIZATION

One exact checkpoint. Exploratory estimates. Decisions within a race are dependent. No cross-model generality claim.

One canonical game, exact state updates.

ENVIRONMENT CONTRACT

VALIDATED ARTIFACT

S **Safe**
progress $+1.0$

U **Unsafe**
progress $+1.5$; risk accumulates

STAGE PAYOFF

$$\pi_i = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}, \qquad q_i = p_{\mathrm{risk}}^{\max} \frac{n_i^U}{T}$$

- simultaneous hidden actions; environment does arithmetic
- minimum 5 rounds, then 20% stop probability
- 100-unit prize; winner/tie-specific terminal setback
- risk caps: 10%, 60%, 90%

Adapted from Fernandez Domingos and Han (2026), arXiv:2607.26034. Human findings are not relabelled as LLM evidence.

Behavior is only the top layer of the audit.

FOUR EVIDENCE LAYERS

01 ENGINE
Exact me-chanics
fixtures, hashes, draws

→

02 ADMISSION
Understand?
rules, state, terminal

→

03 BEHAVIOR
Choose?
live + fixed replay

→

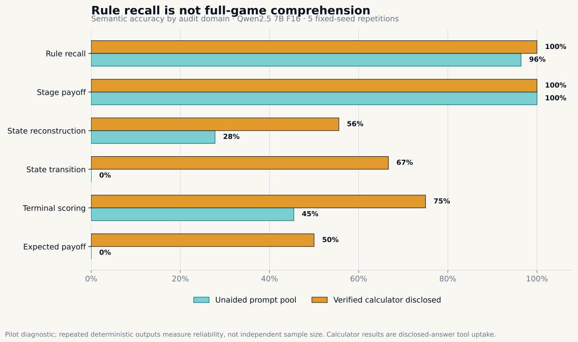
04 MECHANISM
Why?
SAE + causal controls

EACH LAYER HAS ITS OWN PROMOTION GATE

Rule recall did not transfer to game arithmetic.

GAME-UNDERSTANDING AUDIT

EXPLORATORY PILOT



RAW PROBES
685
41 items × prompt conditions

SEMANTIC
59.1%
overall accuracy

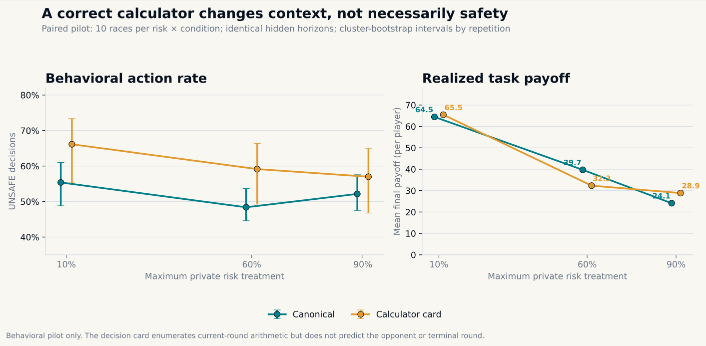
UNAIDED
52.1%
calculator removed

Exact-repeat stability is narrow reliability, not an internal world model. Direct expected-payoff accuracy was 0%.

A calculator changed play; it did not prove understanding.

BEHAVIORAL ABLATION

DIAGNOSTIC ONLY



CANONICAL
52.0%
Unsafe; 558 decisions

CALCULATOR
60.8%
Unsafe; 558 decisions

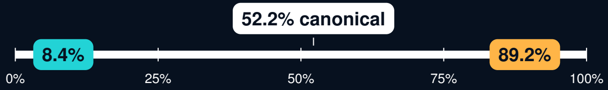
CHANGE
+8.8 pp
30 paired horizon cells

Calculator outputs demonstrate access to disclosed arithmetic. They do not identify internal causal understanding.

Prompt surface moved the same model across 81 points.

18-VARIANT SENSITIVITY GRID

EXPLORATORY PILOT



Extreme whole-trajectory variants

action order reversed: 8.4% Unsafe

risk text near response: 89.2% Unsafe

COVERAGE

540 races

10,044 dependent decisions

PARSE FAILURES

0

all output contracts valid

FIRST ROUND

83.3% flips

emotional framing vs canonical

Whole-trajectory differences include endogenous feedback. First-round flips compare matched initial states and seeds.

Eight stories preserve one mathematical game.

CONTEXT-SKIN INVARIANCE DESIGN

VALIDATED ARTIFACT

CONTROL SKINS

- technology race
- abstract contest

REALISTIC / FICTIONAL PAIRS

- logistics contract / crystal guild
- hospital deployment / colony life support
- robotic expedition / fictional cartography

Payoff-preserving relabeling

Safe=P and Safe=Q crossed
same state, mechanics, model digest

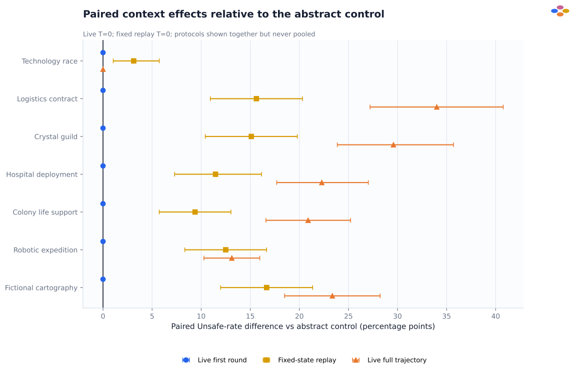
NO SAFE / UNSAFE LABEL LEAK

Primary: 768 live races at temperature 0; 96 engine-reachable states × 8 skins × 2 mappings at temperature 0.

Context effects appeared after feedback began.

PAIRED LIVE AND FIXED-STATE ESTIMATES

DIAGNOSTIC ONLY



FIRST ROUND
0.0 pp
all context contrasts

LIVE MAX
+34.0 pp
logistics vs abstract

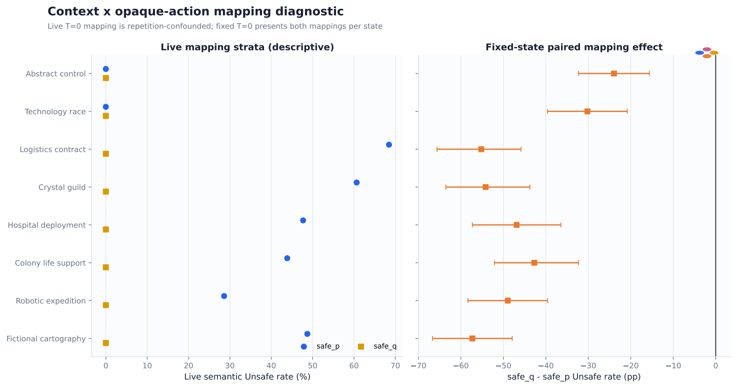
FIXED MAX
+16.7 pp
fictional map vs abstract

Both protocols use T=0. Fixed replay isolates direct prompt effects at sampled states; live trajectories include endogenous feedback. They are shown together but never pooled.

Opaque labels exposed a dominant code-position shortcut.

P/Q MAPPING DIAGNOSTIC

DIAGNOSTIC ONLY



SAFE=P
37.2%
live semantic Unsafe

SAFE=Q
0.0%
live semantic Unsafe

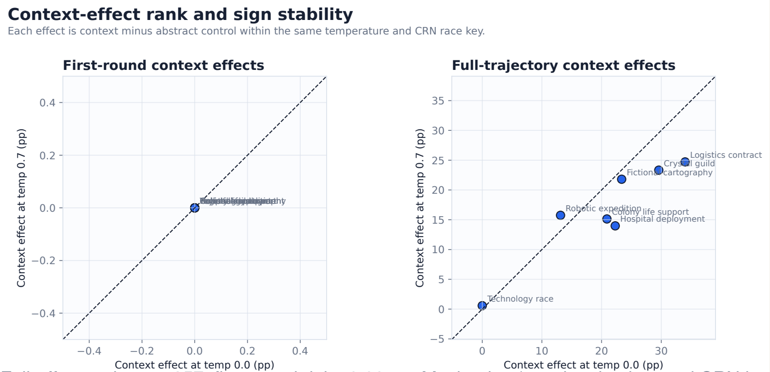
DESIGN FIX
CROSS
mapping within same seed

Live mapping is confounded with repetition-specific horizon; fixed-state replay crosses both mappings within every state and confirms the asymmetry.

Temperature changes trajectories more than first moves.

DECODING ROBUSTNESS: T=0 VERSUS T=0.7

DIAGNOSTIC ONLY



OVERALL

+3.04 pp

95% CI +2.19 to +3.97

ACTIONS

88.2%

paired agreement

TRAJECTORIES

62.6%

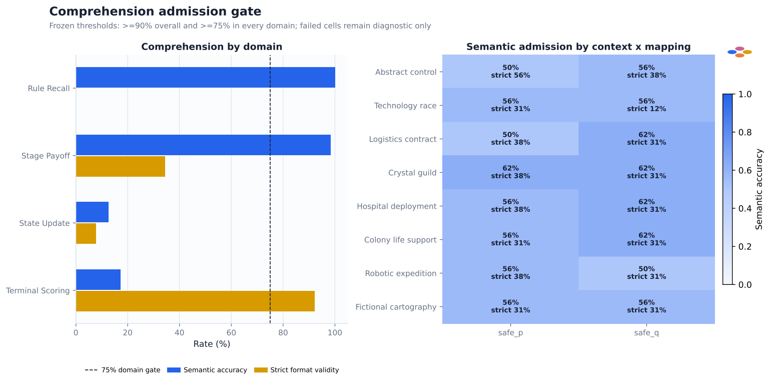
exact player paths

Full-effect rank $\rho = .857$; first-round delta 0.00 pp. Mechanism/template hashes and CRN horizons match, but whole staged-source hashes differ. No comprehension claim.

The context behavior study failed its admission gate.

COMPREHENSION BEFORE INTERPRETATION

DIAGNOSTIC ONLY



RULE RECALL
100%
semantic accuracy

STATE UPDATE
12.5%
semantic accuracy

TERMINAL
17.2%
semantic accuracy

Interpretation: prompt-conditioned action production under identical mechanics - not verified informed optimization.

Recognition audit v1 failed; v2 repaired the contract.

PROTOCOL DEBUGGING IS A RESULT

V1 - REJECTED

0 / 320 admitted

Contract required a named candidate even for generic resemblance. All 960 attempts therefore failed parsing. Raw output was preserved, not rescored.

V2 - ACCEPTED

16 / 16 valid

All responses: generic structural resemblance, high confidence, no named game. Mapping agreement held for all 8 skins.

Self-report neither proves nor rules out memorization, contamination, recognition, or latent understanding.

AUC answers “decodable?” - steering asks “causal?”

ACTUAL-SELF-PLAY FAST-SAE DESIGN

VALIDATED ARTIFACT

SELF-PLAY	DISCOVER	HOLD OUT	STEER	CONTROLS
18 races	12 races	6 races	24 states	matched random
300 decisions	mine SAE features	association only	replay exact	unrelated / sign / recon

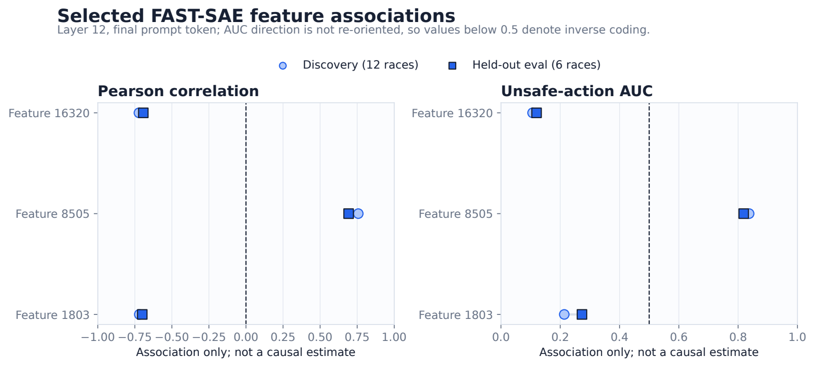
Exact full-sequence likelihood: `ACTION: SAFE` vs `ACTION: UNSAFE`; intervention at layer 12, final prompt token.

Native Qwen2.5-7B-Instruct revision and FAST-SAE revision are pinned in the manifest.

Three SAE features predicted held-out actions.

ASSOCIATION STAGE

DIAGNOSTIC ONLY



F8505
0.819
held-out action AUC

F16320
0.120
inverse-coded AUC

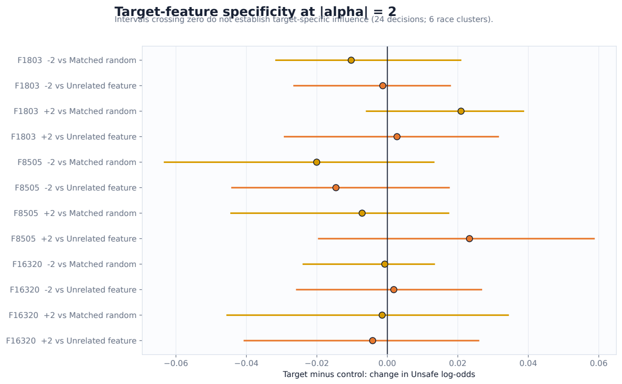
F1803
0.273
inverse-coded AUC

Held-out absolute correlations were 0.692–0.700. Association is useful for screening; it is not a neuron-level reason.

Target steering did not beat its controls.

FIXED-STATE CAUSAL AUDIT

DIAGNOSTIC ONLY



CONTRASTS
0 / 12
CIs exclude zero

RECON FLIPS
12.5%
SAE reconstruction not neutral

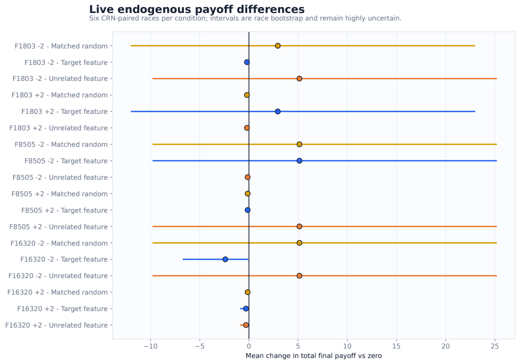
EVAL CLUSTERS
6
race-limited uncertainty

No feature-specific causal-control claim. Dose curves were neither consistently monotone nor sign-reversing.

Live steering changes trajectories, then feedback takes over.

ENDOGENOUS SELF-PLAY EFFECTS

DIAGNOSTIC ONLY



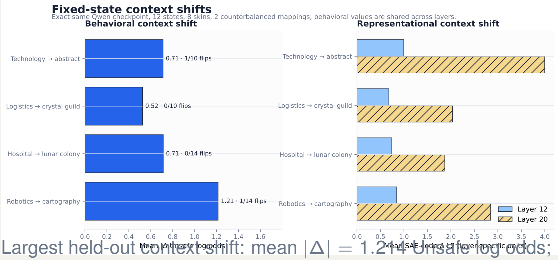
- 114 live trajectories over 6 common-random-number seeds
- direct comparable flips: 1.1–3.9%
- target/control action sequences matched 68.1% on average
- payoff deltas sit on the same scale as controls

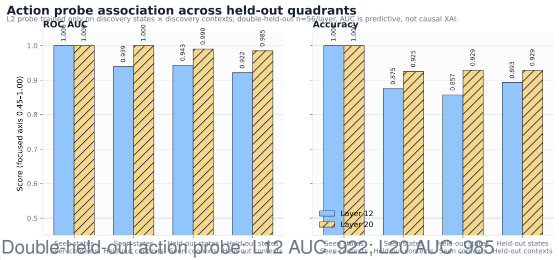
After the first action divergence, changed history and state make later differences total trajectory effects, not fixed-state causal effects.

Context is decodable at two layers - causality still fails.

CONTEXT FAST-SAE SMOKE

DIAGNOSTIC ONLY





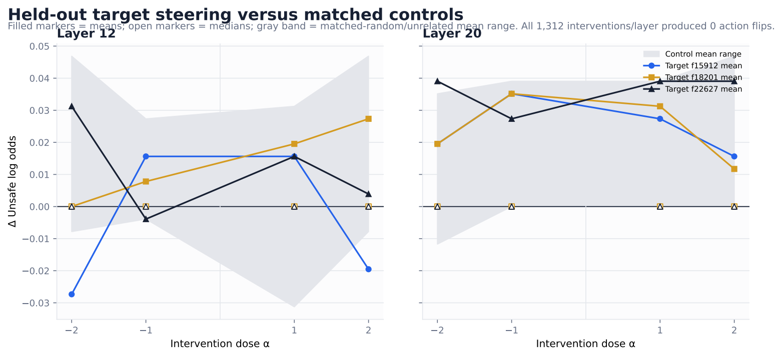
Largest held-out context shift: mean $|\Delta| = 1.214$ unsafe log odds; only 1 discrete flip / 28 pairs.

Double-held-out action probe L12 AUC .922; L20 AUC .985 ($n = 56$).

Layer 20 wins the screen, not the causal claim.

CONTEXT-SAE PROMOTION DECISION

DIAGNOSTIC ONLY



LAYER 20

PROMOTE

capture + analysis

LAYER 12

HOLD

smoke robustness point

STEERING

STOP

0 held-out flips

Require at least 10 discovery flips before steering, or preregister continuous log-odds feature selection.

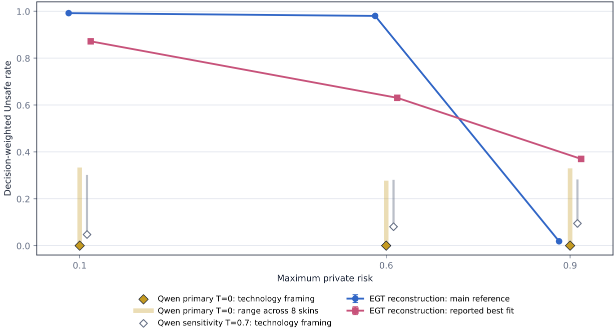
Evolution predicts a risk phase change; Qwen does not.

REDUCED EGT RECONSTRUCTION VERSUS LLM SELF-PLAY

VALIDATED ARTIFACT

Theory and LLM-agent behaviour use the same game mechanics

LLM points are descriptive pilot outputs, not samples from the evolutionary population process



RISK 1 - AU

99.2%

predicted Unsafe

RISK 6 - CAS

98.0%

predicted Unsafe

RISK 9 - CS

1.9%

predicted Unsafe

Faithful, not bitwise: author code/seeds are unavailable. Official EGTTools source parity passed (transition max error 1.11×10^{-16}). LLM rates remain context-dependent and are not evolutionary samples.

What the pilots support - and what they do not.

EVIDENCE LEDGER

SUPPORTED BY THESE PILOTS

- **Engine:** exact fixtures, hashes, complete logs
- **Comprehension:** failures localized to state and terminal arithmetic
- **Behavior:** large prompt, context, and mapping sensitivity
- **SAE:** held-out decodability; scores and actions can be perturbed
- **Recognition:** no named game in the v2 self-report

NOT SUPPORTED

- external validity of the idealized game
- informed utility optimization
- stable population or cross-model effects
- a semantic “Unsafe feature” or target-specific causal control
- absence of contamination, memorization, or latent recognition

All behavioral estimates remain exploratory: comprehension admission failed and only one model family is covered.

Promotion requires stronger controls, not more rows alone.

NEXT CONFIRMATORY GRID

- pass rule, state-update, terminal, and expected-payoff gates in every context $\times$ mapping cell
- cross mapping within identical live race seeds
- at least three model families and fixed decoding
- replay logged states to separate direct effects from feedback
- freeze layer, feature, sign, alpha, and controls before evaluation
- use multiple matched-random directions as an empirical null
- require neutral reconstruction / zero-hook gates
- report race-level, seat-level, and strategy-level estimands together

PRIMARY UNIT OF UNCERTAINTY: WHOLE RACE / MATCHED STATE - NOT DECISION ROW

BOTTOM LINE

Same payoff matrix.
Different context.
Different output.

But the model did not pass the gate required to call that behavior informed optimization.

AUDIT THE GAME → CROSS THE WORDING → HOLD OUT STATES → CONTROL THE CAUSE

Reproducible artifacts in `results/open_source/` • **vector figures embedded directly**

References and reproducibility.

SELECTED SOURCES

VALIDATED ARTIFACT

- Fernandez Domingos, E. and Han, T. A. (2026). *Falling Behind Drives Unsafe Development in an Idealised AI Race Experiment*. arXiv:2607.26034.
- *Same Game, Different Story* (2026). Payoff-preserving contextual robustness for LLM game behavior. arXiv:2607.19670.
- *Framing the Game* (2025). Context effects in game-theoretic LLM evaluation. arXiv:2503.04840.
- *RouteSAE* (2025). Sparse autoencoders for interpreting mixture-of-experts routing. arXiv:2503.08200.


```
model digest:  59805ce4...ff16c
live context:  768 races / 13,680 decisions
fixed replay:  96 states / 1,536 cells
FAST-SAE self-play:  18 races / 300 decisions / 114 steered trajectories
```

Exact configs, raw responses, manifests, hashes, CSV summaries, PNGs, and vector PDFs are stored beside each analysis report.